



## Chapter Four

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# Hyper Text Markup Language (HTML)



# Outline

- ✓ Introduction to HTML
- ✓ HTML tags
  - Basic HTML tags
- ✓ HTML Table
- ✓ HTML Frames
- ✓ HTML Form and Form Controls
- ✓ Inserting Multimedia in HTML
  - Embed vs Video and Audio tags
- ✓ HTML Graphics
  - HTML Canvas
  - HTML SVG

# Introduction to HTML

- ✓ WWW – uses hypertext to cross-reference or link related resources anywhere on the Internet.
- ✓ HTML (HyperText Markup Language) is used to structure and format the content of websites on the World Wide Web.
- ✓ The primary building block to create and structure website content.
  - **HyperText** is a way of organizing text that allows the reader to easily navigate and access related information.
  - **A markup language** is used to add structure and formatting to a text document.
    - not designed for computational tasks but for describing the presentation and structure of documents.
    - interpreted by software applications like web browsers or specialized programs, to render the content according to the defined structure and formatting.
    - Examples: HTML, XML, MathML, LaTeX, etc.

## ...cont'd

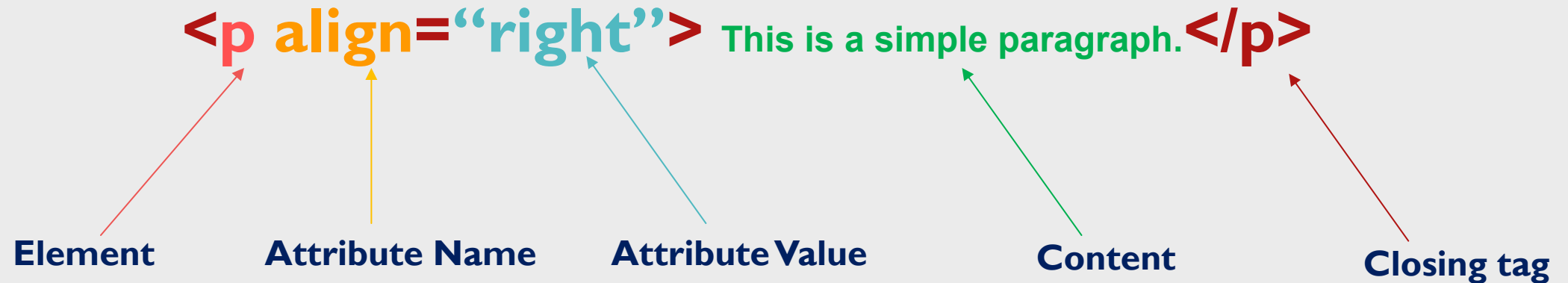
- ✓ It supports a wide range of media types, including **text**, **images**, **audio**, and **video**, which makes web pages more engaging and interactive.
- ✓ HTML is a flexible language that can be used with other technologies, such as CSS and JavaScript, to add additional features and functionality to a web page
- ✓ Since HTML is compatible with all browsers, web pages created in HTML are displayed across a variety of platforms and devices.
- ✓ Furthermore, it is an open and standardized language, which is constantly being updated and improved by a community of developers and experts

# HTML tags

- ✓ HTML tags provide semantic meaning and machine-readability to the content in the page.
- ✓ An element usually consists of an opening tag (`<element_name>`), a closing tag (`</element_name>`), which contain the element's name surrounded by angle brackets, and the content in between:
  - ❑ `<element_name>...content... </element_name>`
- ✓ There are some HTML elements that don't have a closing tag or any contents. These are called void elements.
  - ❑ Void elements include `<img>`, `<meta>`, `<link>` and `<input>`.
- ✓ Element names can be thought of as descriptive keywords for the content they contain, such as video, audio, table, footer.

## ...cont'd

- ✓ A HTML page may consist of potentially hundreds of elements which are then read by a web browser, interpreted and rendered into human readable or audible content on the screen.



### `<p>` opening tag

Represents a common paragraph of text.

## ...cont'd

### ✓ Basic types of commands:

- ☐ Structural tags (mandatory)
- ☐ Heading tags (optional)
- ☐ Formatting tags (optional)
- ☐ Separator tags (optional)
- ☐ List tags (optional)
- ☐ Link tags (optional)

# HTML Structure

- **Structural tags** – these are at the beginning and end of an HTML file.

`<!DOCTYPE html>` Declares the HTML version to be used (HTML5 in this case)

`<html lang="en">` root element of an HTML page which encloses all other elements in the HTML

└─ Specifies the language of the document (e.g., English)

`<head>` Contains meta-information about the HTML document

`<meta charset="UTF-8">`

- `<meta>` tags provide metadata about an HTML document, meaning they don't directly display content on the page but provide supplementary information for browsers or search engines
- `charset="UTF-8"` Specifies the character encoding for the document (UTF-8 for universal character support).

`<meta name="viewport" content="width=device-width, initial-scale=1.0">` Sets the viewport properties for responsive design.

`<title>Document Title</title>` Defines the title of the document displayed in the browser tab.

`</head>`

`<body>` contains the actual content of the HTML document.

`<!-- Content goes here -->`

`</body>`

`</html>`

Elements (tags) are nested one inside another:  
`<html> <head></head> <body></body> </html>`

## ...cont'd

- ✓ **Heading tags** – there are 6 levels of Headings. Level 1 is the largest font size.

Defined with `<h1>` to `<h6>` tags

`<H1> </H1>` Heading level 1

`<H2> </H2>` Heading level 2

`<H3> </H3>` Heading level 3

`<H4> </H4>` Heading level 4

`<H5> </H5>` Heading level 5

`<H6> </H6>` Heading level 6

- ✓ Example: `<H2> This is My Personal Page </H2>`

**Heading 1**

**Heading 2**

**Heading 3**

**Heading 4**

**Heading 5**

**Heading 6**

## ...cont'd

- ✓ **Formatting tags** – are used to control the visual presentation of text content on a web page

- **Highlighting**

- highlight text in a document "due to its relevance in another context"
- `<mark> </mark>`

- **Bold, Italic, and Underline**

- To bold text, use the `<strong>` or `<b>` tags:
  - ✦ `<strong>` is used to indicate that the text is fundamentally or semantically important to the surrounding text, while `<b>` indicates no such importance and simply represents text that should be bolded.
- To italicize text, use the `<em>` or `<i>` tags:
  - ✦ `<em>` is used to indicate that the text should have extra emphasis that should be stressed, while `<i>` simply represents text which should be set off from the normal text around it.
- To underlined text, use the `<U> </U>` tag

## ...cont'd

### ❑ Abbreviation

- To mark some expression as an abbreviation, use `<abbr>` tag:

➤ `<p>I like to write <abbr title="Hypertext Markup Language" >HTML</abbr>! </p>`

### ❑ Inserted, Deleted, or Stricken

- To mark text as inserted, use the `<ins>` tag: `<ins>New Text</ins>`
- To mark text as deleted, use the `<del>` tag: `<del>Deleted Text</del>`
- To strike through text, use the `<s>` or `<strike>` tag: `<strike>strikethrough</strike>`

### ❑ Superscript and Subscript

- To create superscript: `<sup>superscript here</sup>`
- To create subscript: `<sub>subscript here</sub>`

still relevant

### ❑ Larger, Smaller Text

- `<big>Big Text</big>`
- `<small>Small Text </small>`

## ...cont'd

### □ HTML – Fonts

- **Fonts** play a very important role in making a website more user friendly and increasing content readability.
- **Font face and color** depends entirely on the computer and browser that is being used to view your page but you can use HTML `<font>` tag to add style, size, and color to the text on your website.
- You can use a `<basefont>` tag to set all of your text to the same **size, face, and color**.
- The font tag is having three attributes called size, color, and face to customize your fonts.
- **Font Size**
  - Absolute Values (1-7)
    - The default size of a font is 3.
  - Relative Values
    - `<font size = "-1">Text </font>`

## ...cont'd

### ➤ Font Face

- You can set font face using face attribute but be aware that if the user viewing the page doesn't have the font installed, they will not be able to see it.

- Instead user will see the default font face applicable to the user's computer.

```
<font face = "Times New Roman" size = "5">Text</font>
```

- Specify alternate font faces

```
<font face = "Lucida Calligraphy, Comic Sans MS, Lucida Console">
```

### ➤ Font Color

- specify the color that you want by either the color name or hexadecimal code for that color.

```
<font color = "#FF00FF">This text is in pink</font><br />
```

```
<font color = "red">This text is red</font>
```

### ➤ The <basefont> Element

- supposed to set a default font size, color, and typeface for any parts of the document that are not otherwise contained within a <font> tag.

- You can use the <font> elements to override the <basefont> settings.

```
<basefont face = "arial, verdana, sans-serif" size = "2" color = "#ff0000">
```

## ...cont'd

### □ Alignment – some elements have attributes for alignment (ALIGN)

- Three alignment values are : LEFT, RIGHT, CENTER.

<CENTER></CENTER> will center elements

### □ Special characters & symbols

- recognized in HTML as they begin with an ampersand and end with a semi-colon E.g. &value;

- **value** – either be an entity name or a standard ASCII character number

| Special Character | Entity Name | Special Character  | Entity Name   |
|-------------------|-------------|--------------------|---------------|
| Ampersand         | &amp; &     | Greater-than sign  | &gt; >        |
| Asterisk          | &lowast; ** | Less-than sign     | &lt; <        |
| Cent sign         | &cent; ¢    | Non-breaking space | &nbsp; &nbsp; |
| Copyright         | &copy; ©    | Quotation mark     | &quot; "      |
| Fraction one qtr  | &frac14; ¼  | Registration mark  | &reg; ®       |
| Fraction one half | &frac12; ½  | Trademark sign     | &trade; ™     |

## ...cont'd

| Symbol Name               | HTML Entity | Symbol |
|---------------------------|-------------|--------|
| Copyright Sign            | &copy;      | ©      |
| Registered Trademark Sign | &reg;       | ®      |
| Trademark Sign            | &trade;     | ™      |
| Less Than                 | &lt;        | <      |
| Greater Than              | &gt;        | >      |
| Ampersand                 | &amp;       | &      |
| Non-breaking Space        | &nbsp;      |        |
| Em Dash                   | &mdash;     | —      |
| Quotation Mark            | &quot;      | "      |
| Euro                      | &#8364;     | €      |
| British Pound             | &pound;     | £      |
| Japanese Yen              | &yen;       | ¥      |

| Symbol Name               | HTML Entity | Symbol |
|---------------------------|-------------|--------|
| Exclamation Point         | &#33;       | ©      |
| Number Sign               | &#35;       | #      |
| Dollar Sign               | &#36;       | \$     |
| Percent Sign              | &#37;       | %      |
| Single Quote              | &#39;       | '      |
| Selection Sign            | &sect;      | §      |
| Not Sign                  | &not;       | ¬      |
| Cent Sign                 | &cent;      | —      |
| Fraction three quarters   | &frac34;    | "      |
| Micro Sign                | &micro;     | μ      |
| Superscript two - squared | &sup2;      | ²      |
| Degree Sign               | &deg;       | °      |

...cont'd

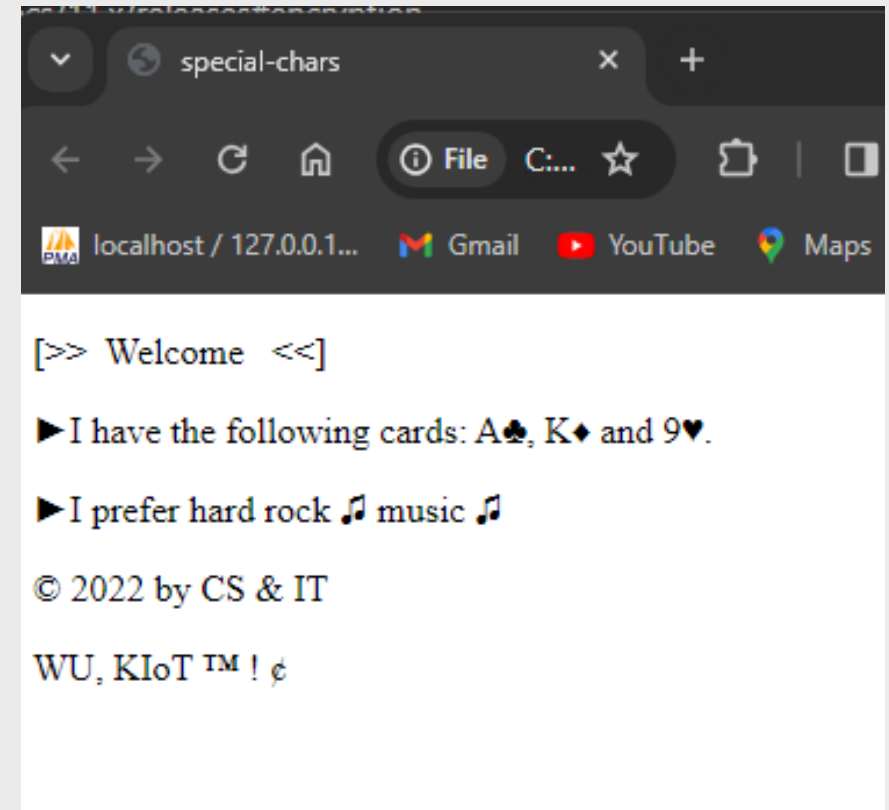
[illegible]

I have the following cards:  
A, K; and 9.

<p>#9658;l prefer hard rock #9835;  
music #9835;</p>

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WU, KIoT &trade; &#33; &cent;



## ...cont'd

- ✓ **Separators tags** – these tags separate words, or sentences on a page.
  - ❑ Paragraph (`<p>`)
    - Paragraph elements are typically used to represent paragraphs of text, but they can also be styled to create vertical spacing or visual separation between blocks of content.
  - ❑ Line Break (`<br>`):
    - `<br />` Break; breaks up text onto two lines
  - ❑ Horizontal Rule (`<hr>`)
    - `<hr />` Make a Horizontal Rule (or line)

## ...cont'd

### ✓ Lists Tag

□ HTML offers three ways for specifying lists:

1. **Ordered lists** use ordinal sequences to indicate the order of list elements,

➤ list of numbered items

<OL TYPE="i" START="3"> specify a starting number for an ordered list.

| TYPE | Numbering      | Styles            |
|------|----------------|-------------------|
| I    | Arabic numbers | 1,2,3, .....      |
| a    | Lower alpha    | a, b, c, .....    |
| A    | Upper alpha    | A, B, C, .....    |
| i    | Lower roman    | i, ii, iii, ..... |
| I    | Upper roman    | I, II, III, ..... |

## ...cont'd

2. **Unordered lists** use a defined symbol such as a bullet to list elements in no designated order, and

- list of unnumbered items (bullet)
- Browsers will usually change list mark in nested lists.
- Three bullet types: disc(default), circle, square.

`<ul type = "square">`

```
<ul type = "square">
  <li>Beetroot</li>
  <li>Ginger</li>
  <li>Potato</li>
  <li>Radish</li>
</ul>
```

- Beetroot
- Ginger
- Potato
- Radish

3. **Description lists** use indents to list elements with their children.

- each item in a DL consists of one or more Definition Terms (DT elements), followed by one or more Definition Description (DD elements).

```
<DL>
<DT> HTML </DT>
<DD> Hyper Text Markup Language </DD>
<DT> DOG </DT>
<DD> A human's best friend!</DD>
</DL>
```

```
HTML
  Hyper Text Markup Language
DOG
  A human's best friend!
```

## ...cont'd

### ✓ Link tags

- ❑ A webpage can contain various links that take you directly to other pages and even specific parts of a given page. These links are known as hyperlinks.
- ❑ **Hyperlinks** allow visitors to navigate between Web sites by clicking on words, phrases, and images.

- The `<a>` tag (anchor tag) creates a clickable link.

`<a href = "URL" target= "attributes-list" title= "title">`

Link Text `</a>`

the visible text that users will see and click on.

add optional attributes to further customize the link behavior (e.g., `target="_blank"` to open the link in a new tab).

- ❑ The target attribute can have one of the following values:
  - `_self` - Default. Opens the document in the same window/tab as it was clicked
  - `_blank` - Opens the document in a new window or tab

## ...cont'd

- ❑ same server in the same directory:
  - `<a href="form.html">Fill Our Form</a>`
- ❑ same server in the parent directory:
  - `<a href="../parent.html">Parent</a>`
- ❑ same server in the subdirectory stuff:
  - `<a href="stuff/cat.html">Catalog</a>`
- ❑ Link to an external Web site
  - `<a href="https://www.w3schools.com/html/" target="_blank"> w3schools tutorial</a>`
    - Always use a absolute URL
    - Using `target="_blank"` attribute opens link in new window

## ...cont'd

### ❑ Absolute URLs

- Contain all the information: An absolute URL includes all the necessary details to locate a resource, including:
- Protocol (e.g., http://, https://) `<a href="https://www.google.com/">Google</a>`
- Domain name (e.g., www.example.com)
- Path to the specific file or resource (e.g., /images/hikingboots.jpg)

### ❑ Relative URLs

- Shorter and more concise: Relative URLs provide the location of a resource relative to the current web page.
- Base URL needed: They rely on a base URL (the URL of the current page) to determine the complete location.
- A local link (a link to a page within the same website) is specified with a relative URL (without the "https://www" part): `<a href="html_images.asp">HTML Images</a>`

## ...cont'd

### ❑ Use an Image as a Link

```
<a href="default.asp">  
  
</a>
```

### ❑ Email Links

- The <a> tag with href="mailto:email@example.com" creates a link that opens a user's email client to send an email.

```
<a href = "mailto: abc@example.com">Send Email</a>
```

- You can specify a default email subject and email body along with your email address. Following is the example to use default subject and body.

```
<a href = "mailto:abc@example.com?subject = Feedback&body = Message">  
Send Feedback </a>
```

### ❑ Link that dials a number

```
<a href="tel:11234567890" >Call us</a>
```

## ...cont'd

### □ Bookmark

- Bookmarks in HTML allow users to jump to specific sections within a webpage.
- Creating the Bookmark (Anchor):
  - ✚ Use the `<a>` tag with an `id` attribute to define the location of the bookmark.
  - ✚ Place this tag at the point in your HTML where you want users to be able to jump to.
- Linking to the Bookmark:
  - ✚ Use another `<a>` tag with the `href` attribute set to `#` followed by the `id` you assigned to the bookmark.
  - ✚ This link text becomes what users click on to jump to the bookmarked section.

```
<ul>  
  <li><a href="#chapter1">Chapter 1: Introduction</a></li>  
  <li><a href="#chapter2">Chapter 2: Getting Started</a></li>  
</ul>
```

```
<h2 id="chapter1">Chapter 1: Introduction</h2>  
<h2 id="chapter2">Chapter 2: Getting Started</h2>
```

## ...cont'd

### ✓ Meta Tags

- ❑ Describing things like **the document's character encoding, description, keywords, author, viewport size for mobile devices**, and more.
- ❑ Communicate with search engines and browsers, influencing how a webpage is indexed and displayed.
  - Search Engines:
    - ☛ Indexing: show your webpage in search results
    - ☛ Ranking: understand its content for ranking (relevance)
  - Browsers:
    - ☛ Display: ensure proper layout on phones and tablets.
    - ☛ Character Encoding: prevent garbled text by specifying the correct character set.
- ❑ Does not have a closing tag but it carries information within its attributes.
- ❑ You can include one or more meta tags in your document
- ❑ Do not impact physical appearance of the document

## ...cont'd

<head>

<title>Meta Tags Example</title>

<meta name = "keywords" content = "HTML, Meta Tags, Metadata" />

<meta http-equiv = "Content-Type" content = "text/html; charset = UTF-8" />

<meta http-equiv = "refresh" content = "5; url = http://www.tutorialspoint.com" />

</head>

<meta http-equiv = "refresh" content = "5; url = http://www.tutorialspoint.com" />

- used to refresh the page or to set a cookie.
- Values include content-type, expires, refresh and set-cookie.

redirecting current page to another page after 5 seconds.

### Setting Cookies

- Cookies are data, stored in small text files on your computer and it is exchanged between web browser and web server to keep track of various information based on your web application need.
- You can use <meta> tag to store cookies on client side and later this information can be used by the Web Server to track a site visitor.

<meta http-equiv = "cookie" content = "userid = xyz; expires = Wednesday, 08-Aug-15 23:59:59 GMT;" />

## ...cont'd

### ✓ HTML image

- ❑ HTML images are defined with the `<img>` tag.
- ❑ ``
- ❑ ``
- ❑ Image attributes:

|        |                                                  |
|--------|--------------------------------------------------|
| src    | specifies the path (URL) to the image file       |
| alt    | alternative text for display (e.g. in text mode) |
| height | Number of pixels of the height                   |
| width  | Number of pixels of the width                    |
| border | Size of border, 0 for no border                  |

## ...cont'd

### □ HTML Image Map

- An HTML image map is a way to turn specific sections of an image into clickable links.
- HTML provides the `<img>` tag to insert images and the `<map>` tag to define clickable areas on that image.
- The `<map>` tag is linked to the `<img>` tag using the `usemap` attribute, specifying which image map corresponds to that specific image.
- Image maps can add a layer of interactivity to your webpages and enhance user experience by allowing them to click on specific parts of an image for more information.

```

<map name="worldmap">
  <area shape="rect" coords="100,50,200,100" href="europe.html" alt="Link to Europe page">
  <area shape="circle" coords="300,150,40" href="asia.html" alt="Link to Asia page">
</map>
```

## ...cont'd

### ✓ HTML Comments

- ❑ are lines of code hidden from the web browser but included in the source code for programmers or web developers.
- ❑ They provide annotations, reminders, or explanations within the HTML document to improve readability and maintainability.
- ❑ HTML comments are placed in between `<!-- ... -->` tags.
- ❑ So, any content placed with-in `<!-- ... -->` tags will be treated as comment and will be completely ignored by the browser.

# HTML Table

- ✓ The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.
- ✓ The HTML tables are created using the `<table>` tag in which the `<tr>` tag is used to create table rows and `<td>` tag is used to create data cells.
- ✓ The elements under `<td>` are regular and left aligned by default

```
<table border = "1">
  <tr>
    <th>Name</th>
    <th>Salary</th>
  </tr>
  <tr>
    <td>Row 1, Column 1</td>
    <td>Row 1, Column 2</td>
  </tr>
  <tr>
    <td>Row 2, Column 1</td>
    <td>Row 2, Column 2</td>
  </tr>
</table>
```

...cont'd

## ✓ Cellpadding and Cellspacing Attributes

```
<table border = "1" cellpadding = "5" cellspacing = "5">
  <tr>
    <th>Name</th>
    <th>Salary</th>
  </tr>
  <tr>
    <td>Ramesh Raman</td>
    <td>5000</td>
  </tr>
  <tr>
    <td>Shabbir Hussein</td>
    <td>7000</td>
  </tr>
</table>
```

| Name            | Salary |
|-----------------|--------|
| Ramesh Raman    | 5000   |
| Shabbir Hussein | 7000   |

## ...cont'd

### □ Colspan and Rowspan Attributes

- You will use colspan attribute if you want to merge two or more columns into a single column.
- Similar way you will use rowspan if you want to merge two or more rows.

```
<table border = "1">
  <tr>
    <th>Column 1</th>
    <th>Column 2</th>
    <th>Column 3</th>
  </tr>
  <tr>
    <td rowspan = "2">Row 1 Cell 1</td>
    <td>Row 1 Cell 2</td>
    <td>Row 1 Cell 3</td>
  </tr>
  <tr>
    <td>Row 2 Cell 2</td>
    <td>Row 2 Cell 3</td>
  </tr>
  <tr>
    <td colspan = "3">Row 3 Cell 1</td>
  </tr>
</table>
```

| Column 1     | Column 2     | Column 3     |
|--------------|--------------|--------------|
| Row 1 Cell 1 | Row 1 Cell 2 | Row 1 Cell 3 |
|              | Row 2 Cell 2 | Row 2 Cell 3 |
| Row 3 Cell 1 |              |              |

## ...cont'd

### □ Tables Backgrounds

- You can set table background using one of the following two ways –
  - bgcolor attribute – You can set background color for whole table or just for one cell.
  - background attribute – You can set background image for whole table or just for one cell.

Note – The bgcolor, background, and bordercolor attributes deprecated in HTML5.  
Do not use these attributes.

```
<table border = "1" bordercolor = "green" bgcolor = "yellow">
```

```
<table border = "1" bordercolor = "green" background = "/images/test.png">
```

### □ Table Height and Width

- set a table width and height using width and height attributes in terms of pixels or in terms of percentage of available screen area.

```
<table border = "1" width = "400" height = "150">
```

## ...cont'd

### ❑ Table Caption

- The caption tag will serve as a title or explanation for the table and it shows up at the top of the table. This tag is deprecated in newer version of HTML/XHTML.

```
<table border = "1" width = "100%">  
  <caption>This is the caption</caption>
```

### ❑ Table Header, Body, and Footer

- Tables can be divided into three portions – a header, a body, and a foot.

```
<thead> – to create a separate table header.  
<tbody> – to indicate the main body of the table.  
<tfoot> – to create a separate table foote
```

### ❑ Nested Tables

- You can use one table inside another table. Not only tables you can use almost all the tags inside table data tag <td>.

## ...cont'd

- HTML layout example is achieved using a table with 3 rows and 2 columns but the header and footer column spans both columns using the colspan attribute –

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Layout using Tables</title>
  </head>
  <body>
    <table width = "100%" border = "0">
      <tr>
        <td colspan = "2" bgcolor = "#b5dcb3">
          <h1>This is Web Page Main title</h1>
        </td>
      </tr>
      <tr valign = "top">
        <td bgcolor = "#aaa" width = "50">
          <b>Main Menu</b><br />
          HTML<br />
          PHP<br />
          PERL...
        </td>
        <td bgcolor = "#eee" width = "100" height = "200">
          Technical and Managerial Tutorials
        </td>
      </tr>
    </table>
```

```
<td colspan = "2" bgcolor = "#b5dcb3">
  <center>
    Copyright © 2007 Tutorialspoint.com
  </center>
</td>
</tr>
</table>
</body>
</html>
```

|                                            |                                    |
|--------------------------------------------|------------------------------------|
| <b>This is Web Page Main title</b>         |                                    |
| <b>Main Menu</b><br>HTML<br>PHP<br>PERL... | Technical and Managerial Tutorials |

# HTML Frames

- ✓ HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document.
- ✓ A collection of frames in the browser window is known as a frameset.
- ✓ The window is divided into frames in a similar way the tables are organized: into rows and columns.

## ❑ Disadvantages of Frames

- There are few drawbacks with using frames, so it's never recommended to use frames in your webpages –
- Some smaller devices cannot cope with frames often because their screen is not big enough to be divided up.
- Sometimes your page will be displayed differently on different computers due to different screen resolution.
- The browser's back button might not work as the user hopes.
- There are still few browsers that do not support frame technology.

Note – The `<frame>` tag deprecated in HTML5. Do not use this element.

## ...cont'd

- ✓ Frames are found in Framesets
- ✓ Framesets define the 'layout' of the frames it contains
- ✓ Framesets can be nested in one another to provide a more complex layout.

### □ Creating Frames

Tag format:

Single frameset

```
<frameset>
  <frame>
  <frame>
  ...
</frameset>
```

Nested frameset

```
<frameset>
  <frame>
  <frameset>
    <frame>
    <frame>
    ...
  </frameset>
  <frame>
  ...
</frameset>
```

## ...cont'd

### ✓ Frameset Attributes

- ❑ rows = “row dimensions” { ex. rows=“10%, 90%” }
- ❑ cols = “column dimensions” { ex. cols=“20%, \*, 30%” }
- ❑ border = “value”
- ❑ bordercolor = “color”
- ❑ frameborder = “value” { 0, 1, no, yes }
- ❑ framespacing = “value”

### ✓ Frame Attributes

- ❑ name = “frame\_name”
- ❑ src = “url” { url can be local or external }
- ❑ noresize [= “noresize”]
- ❑ scrolling = “value” { auto, yes, no }
- ❑ allowtransparency = “value” { in % }
- ❑ frameborder = “value” { 0, 1, yes, no }
- ❑ bordercolor = “color”

# HTML Iframes

- ✓ Inline frames (iframe) also called floating frames is used to embed a webpage within a webpage.

```
<iframe src="https://www.javatpoint.com/" height="300" width="600"></iframe>
```

- ❑ **The <iframe> Tag Attributes**

- **src:** It is used to specify the URL of the website to be loaded.
- **title:** It is good practice to include a title attribute so that screen readers can read the title to users.
- **width:** set the width of <iframe>.
- **height:** set the height of <iframe>.

- ❑ <iframe> tag defines a rectangular region within the document in which the browser can display a separate document

- ❑ Tag format:

```
<iframe></iframe>
```

## ...cont'd

```
<html>
```

```
  <head>
```

```
    <title>iFrame sample</title>
```

```
  </head>
```

```
  <body>
```

```
    Below is the google site <br>
```

```
    <iframe height="600px" width="600px" src=https://www.google.com scrolling="auto">
```

```
      </iframe>
```

```
  </body>
```

```
</html>
```

## ...cont'd

### ✓ Colors

- ❑ Colors are very important to give a good look and feel to your website.
- ❑ You can specify colors on page level using `<body>` tag or you can set colors for individual tags using `bgcolor` attribute.
- ❑ The `<body>` tag has following attributes which can be used to set different colors –
  - `bgcolor` – sets a color for the background of the page.
  - `text` – sets a color for the body text.
  - `alink` – sets a color for active links or selected links.
  - `link` – sets a color for linked text.
  - `vlink` – sets a color for visited links – that is, for linked text that you have already clicked on.

## ...cont'd

### □ HTML Color Coding Methods

- There are three different methods to set colors in your web page. Color name, Hex Code, and Color decimal or percentage values

- **Color names** – You can specify color names directly like green, blue or red.

- 16 Colors names and it is recommended to use them.

|                                                                                    |        |                                                                                      |       |                                                                                      |        |                                                                                      |         |
|------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------|---------|
|   | Black  |   | Gray  |   | Silver |   | White   |
|   | Yellow |   | Lime  |   | Aqua   |   | Fuchsia |
|   | Red    |   | Green |   | Blue   |   | Purple  |
|  | Maroon |  | Olive |  | Navy   |  | Teal    |

```
<body text = "blue" bgcolor = "green">
```

```
<table bgcolor = "black">
```

```
<font color = "white">This text will appear white on black background.</font>
```

## ...cont'd

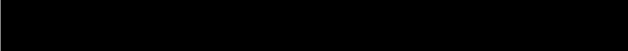
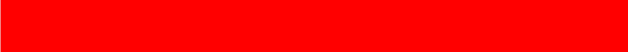
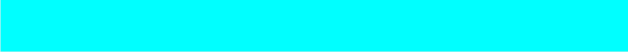
- **Hex codes** – A six-digit code representing the amount of red, green, and blue that makes up the color.
  - A hexadecimal is a 6 digit representation of a color. The first two digits(RR) represent a red value, the next two are a green value(GG), and the last are the blue value(BB).
  - A hexadecimal value can be taken from any graphics software like Adobe Photoshop, Paintshop Pro or MS Paint.
  - Each hexadecimal code will be preceded by a pound or hash sign #.

| Color                                                                                | Color HEX |
|--------------------------------------------------------------------------------------|-----------|
|    | #000000   |
|    | #FF0000   |
|    | #00FF00   |
|   | #0000FF   |
|  | #FFFF00   |
|  | #00FFFF   |
|  | #FF00FF   |
|  | #C0C0C0   |
|  | #FFFFFF   |

```
<body text = "#0000FF" bgcolor = "#00FF00">  
<table bgcolor = "#000000">
```

## ...cont'd

- **Color decimal or percentage values** – This value is specified using the `rgb( )` property.
  - This color value is specified using the `rgb( )` property. This property takes three values, one each for red, green, and blue.
  - The value can be an integer between 0 and 255 or a percentage.
  - **Note** – All the browsers does not support `rgb()` property of color so it is recommended not to use it.

| Color                                                                                | Color RGB                     |
|--------------------------------------------------------------------------------------|-------------------------------|
|    | <code>rgb(0,0,0)</code>       |
|    | <code>rgb(255,0,0)</code>     |
|    | <code>rgb(0,255,0)</code>     |
|    | <code>rgb(0,0,255)</code>     |
|   | <code>rgb(255,255,0)</code>   |
|  | <code>rgb(0,255,255)</code>   |
|  | <code>rgb(255,0,255)</code>   |
|  | <code>rgb(192,192,192)</code> |
|  | <code>rgb(255,255,255)</code> |

`<body text = "rgb(0,0,255)" bgcolor = "rgb(0,255,0)">`

## ...cont'd

### □ Marquee

- An HTML marquee is a scrolling piece of text displayed either horizontally across or vertically down your webpage depending on the settings.
- This is created by using HTML `<marquee>` tag.
- **Note – The `<marquee>` tag deprecated in HTML5. Do not use this element, instead you can use JavaScript and CSS to create such effects.**

```
<marquee attribute_name = "attribute_value"....more attributes>  
    One or more lines or text message or image  
</marquee>
```

#### ➤ The `<marquee>` Tag Attributes

- Width, Height, Direction, scrolldelay, loop, bgcolor,
- Behavior – This specifies the type of scrolling of the marquee. This can have a value like scroll, slide and alternate.

# HTML Form

- ✓ HTML Forms are required, when you want to collect some data from the site visitor.
  - ❑ For example, during user registration you would like to collect information such as name, email address, credit card, etc.
- ✓ A form will take input from the site visitor and then will post it to a back-end application.
- ✓ The back-end application will perform required processing on the passed data based on defined business logic inside the application.
- ✓ There are various form elements available like text fields, textarea fields, drop-down menus, radio buttons, checkboxes, etc.
- ✓ The HTML `<form>` tag is used to create an HTML form and it has following syntax –

```
<form action = "Script URL" method = "GET | POST">  
  form elements like input, textarea etc.  
</form>
```

# ...cont'd

## ✓ Form Attributes

### ❑ action

- Backend script ready to process your passed data.
- Specifies an address (url) where to submit the form (default: the submitting page).

### ❑ method

- Method to be used to upload data. The most frequently used are GET and POST methods.
- Specifies HTTP method used when submitting form (default: GET).

### ❑ target

- Specify the target window or frame where the result of the script will be displayed. It takes values like `_blank`, `_self`, `_parent` etc.

### ❑ enctype

- You can use the enctype attribute to specify how the browser encodes the data before it sends it to the server. It is only applicable if we use the POST method.
- Possible values are –
  - `application/x-www-form-urlencoded` – This is the standard method most forms use in simple scenarios.
  - `multipart/form-data` – This is used when you want to upload binary data in the form of files like image, word file etc.

### ❑ name: Specifies a name used to identify the form

### ❑ novalidate: Specifies that the browser should not validate the form.

# HTML Form Controls

- ✓ There are different types of form controls that you can use to collect data using HTML form
  1. Text Input Controls
  2. Checkboxes Controls
  3. Radio Box Controls
  4. Select Box Controls
  5. File Upload Box
  6. Button Control

## ...cont'd

### 1. Text Input Controls

- ❑ Input tags are HTML elements designed to take a specific form of input from users
- ❑ List of attributes for `<input>` tag for creating text field
  - **type**
    - ☛ Indicates the type of input control and for text input control it will be set to text.
  - **name**
    - ☛ Used to give a name to the control which is sent to the server to be recognized and get the value.
  - **value**
    - ☛ This can be used to provide an initial value inside the control.
  - **size**
    - ☛ Allows to specify the width of the text-input control in terms of characters.
  - **maxlength**
    - ☛ Allows to specify the maximum number of characters a user can enter into the text box.

## ...cont'd

- There are three types of text input used on forms –

1. Single-line text input controls

- This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML `<input>` tag.

```
<form >
```

```
    First name: <input type = "text" name = "first_name" />
```

```
    <br>
```

```
    Last name: <input type = "text" name = "last_name" />
```

```
</form>
```

First name:

Last name:

## ...cont'd

### 2. Password input controls

- This is also a single-line text input but it masks the character as soon as a user enters it.
- They are also created using HTML `<input>` tag but type attribute is set to password.

Password: `<input type = "password" name = "password" />`

### 3. Multiple-Line Text Input Controls

- This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML `<textarea>` tag.
- **attributes for `<textarea>` tag.**
  - ✚ **rows:** Indicates the number of rows of text area box.
  - ✚ **cols:** Indicates the number of columns of text area box

`<form>`

Description : `<br />`

`<textarea rows = "5" cols = "50" name = "description">`

Enter description here...

`</textarea>`

`</form>`

## 2. Checkbox Control

- ❑ used to allow a user to select one or more options from a limited number of options.
- ❑ They are also created using HTML `<input>` tag but type attribute is set to checkbox..

`<form>`

`<input type = "checkbox" name = "maths" value = "on"> Maths`

`<input type = "checkbox" name = "physics" value = "on"> Physics`

`</form>`

- ❑ **attributes for `<checkbox>` tag.**

- **Type:** Indicates the type of input control and for checkbox input control it will be set to checkbox..
- **Name:** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **Value :** The value that will be used if the checkbox is selected.
- **Checked:** Set to checked if you want to select it by default.

...cont'd

### 3. Radio Button Control

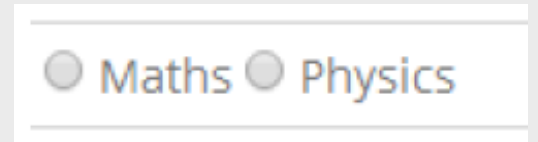
- ❑ used when out of many options, just one option is required to be selected.
- ❑ created using HTML `<input>` tag but type attribute is set to radio.

```
<form>
```

```
<input type="radio" name="subject" value="maths"> Maths
```

```
<input type="radio" name="subject" value="physics"> Physics
```

```
</form>
```



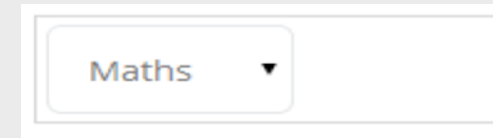
#### ❑ Attribute & Description

- **type:** Indicates the type of input control and for checkbox input control it will be set to a radio.
- **name:** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **value:** The value that will be used if the checkbox is selected
- **checked:** Set to checked if you want to select it by default

## 4. Select Box Controls

- also called drop down box provides option to list down various options in the form of drop down list, from where a user can select one or more options.

```
<form>  
<select name="dropdown">  
<option value="Maths" selected>Maths</option>  
<option value="Physics">Physics</option></select>  
</form>
```



- Following is the list of important attributes of **<select>** tag

- Attribute & Description

- **name:** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **size:** This can be used to present a scrolling list box.
- **multiple:** If set to "multiple" then allows a user to select multiple items from the menu.

## ...cont'd

- ❑ Following is the list of important attributes of `<option>` tag –
  - Attribute & Description
    - **value:** The value that will be used if an option in the select box box is selected.
    - **selected:** Specifies that this option should be the initially selected value when the page loads.
    - **label:** An alternative way of labeling options

## ...cont'd

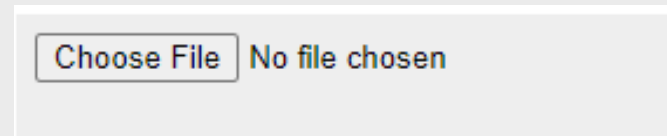
### 5. File Upload Box

- ❑ If you want to allow a user to upload a file to your web site, you will need to use a file upload box, also known as a file select box.
- ❑ This is also created using the `<input>` element but type attribute is set to file.

```
<form>
```

```
  <input type = "file" name = "fileupload" accept = "image/*" />
```

```
</form>
```



- ❑ Following is the list of important attributes of file upload box –
- ❑ Attribute & Description
  - **name:** Used to give a name to the control which is sent to the server to be recognized and get the value.
  - **accept:** Specifies the types of files that the server accepts.

## 6. Button Controls

- ❑ You can also create a clickable button using `<input>` tag by setting its type attribute to button.

- ❑ **Type & Description**

- **submit:** This creates a button that automatically submits a form.
- **reset:** This creates a button that automatically resets form controls to their initial values.
- **button:** This creates a button that is used to trigger a client-side script when the user clicks that button.
- **image:** This creates a clickable button but we can use an image as background of the button.

```
<form>  
  <input type = "submit" name = "submit" value = "Submit" />  
  <input type = "reset" name = "reset" value = "Reset" />  
  <input type = "button" name = "ok" value = "OK" />  
  <input type = "image" name = "imagebutton" src = "/html/images/logo.png" />  
</form>
```

- **name:** The name attribute uniquely identifies a button

- **value:** The value attribute gives a label to the button

## ...cont'd

### ✓ Hidden Controls

- ❑ used to hide data inside the page which later on can be pushed to the server.
- ❑ This control hides inside the code and does not appear on the actual page.

```
<form>  
  <p>This is page 10</p>  
  <input type = "hidden" name = "pagename" value = "10" />  
  <input type = "submit" name = "submit" value = "Submit" />  
  <input type = "reset" name = "reset" value = "Reset" />  
</form>
```

### ✓ The Submit & Reset Button

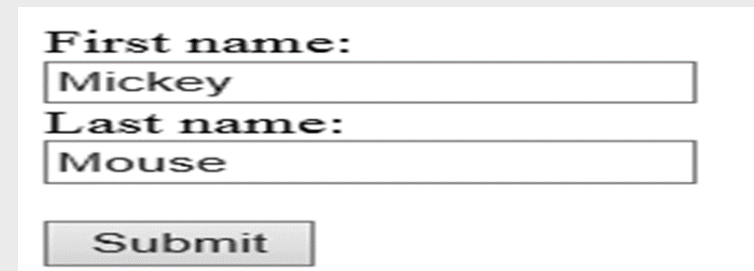
- ❑ Defines a button for submitting a form to a form-handler

`<input type="reset">` – clears all data entered so far

# HTML Form Handlers

- ✓ When a user clicks the submit button, the form needs to send that information somewhere to be processed.
- ✓ This is where the form handler comes in.
  - ❑ The form-handler is specified in the form's **action** attribute:
  - ❑ To be submitted correctly, each input field must have name attribute.

```
<form action="action_page.php">  
  First name: <input type="text" name="firstname" value="Mickey"><br>  
  Last name: <input type="text" name="lastname" value="Mouse">  
  <br><input type="submit" value="Submit">  
</form>
```



First name:  
Mickey  
Last name:  
Mouse  
Submit

## ...cont'd

❑ **Action attribute** – defines the action to be performed when the form is submitted

- The common way to submit a form to a server, is by using a submit button.

```
<form action="action_page.php">
```

- In the example above, server-side script (PHP program) is specified to handle the submitted form (process the form data):
- If action attribute is omitted, the action is set to current page

## ...cont'd

- ✓ HTTP works as a request-response protocol between a client and server.
- ✓ The HTTP protocol allows a variety of mechanisms for sending requests to a server
- ✓ The most commonly used **HTTP request methods** are
  - **GET:**
    - This method is used to retrieve data from a resource.
    - It's the most common method used when browsing websites, as it tells the server to send back the information for the requested resource, like a web page or an image.
    - Less secure because it exposes data in the URL .
    - Can only send a limited amount of data
    - It is good for search content in order to let users to used it again.

```
<form action="action_page.php" method="get">
```

## ...cont'd

### □ POST:

- This method is typically used to submit data to a server.
- For instance, when you fill out a form on a website and click submit, the browser uses a POST request to send the form data to the server for processing (like creating an account or placing an order).  
`<form action="action_page.php" method="post">`
- Offers greater security
- Transmits data within the HTTP request body
- Can handle larger amounts of data
- Better suited for sending sensitive information

### □ PUT: This method is used to update or completely replace a resource with the data sent in the request.

- It's often used with RESTful APIs for specific data manipulation.

# Exercise

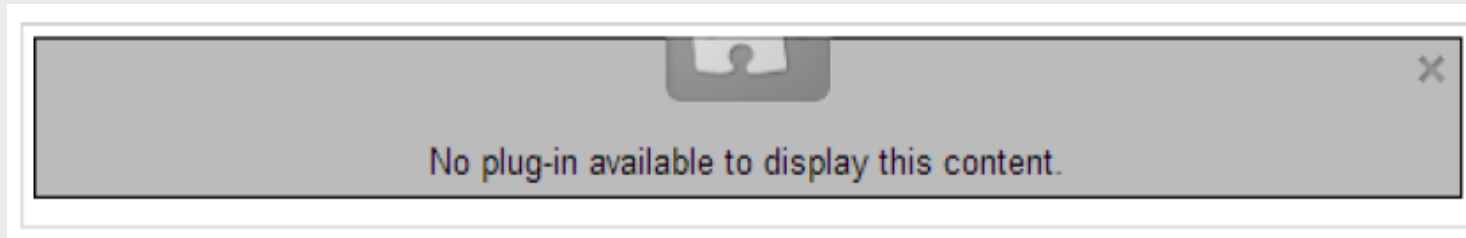
- ❑ Create an HTML page which displays student registration screen with
  - ☛ a **name** field //mandatory and has some description
  - ☛ an **ID** field // mandatory and has some description
  - ☛ a **department** field {select box with values:
    - 1. Computer Science //default
    - 2. Information Technology
  - ☛ a **semester** field {radio with values:
    - 1. I //default
    - 2. II
  - ☛ an **academic year** field {select box with values: 1998-2000, 1999 default}
  - ☛ a button to reset the form
  - ☛ a button to submit the form

# HTML Embedded Multimedia

- ✓ Sometimes you need to add music or video into your web page using HTML tag called `<embed>`
  - ❑ This tag causes the browser itself to include controls for the multimedia automatically provided browser supports `<embed>` tag and given media type.
  - ❑ You can also include a `<noembed>` tag for the browsers which don't recognize the `<embed>` tag
- ✓ Example – to play an embedded midi file:

- ❑ A MIDI file (short for Musical Instrument Digital Interface) is a type of computer file that contains instructions for playing music, rather than the actual audio data itself.

```
<embed src= " /html/yourfile.mid" width="100%" height="60" >  
</embed>
```



## ...cont'd

- ❑ The <embedded> Tag Attributes for radio button
  - **Align:** Determines how to align the object. It can be set to either center, left or right.
  - **Autostart:** This boolean attribute indicates if the media should start automatically. You can set it either true or false.
  - **Loop:** Specifies if the sound should be played continuously (set loop to true), a certain number of times (a positive value) or not at all (false)
  - **Playcount:** Specifies the number of times to play the sound. This is alternate option for loop if you are using IE.
  - **Hidden:** Specifies if the multimedia object should be shown on the page. A false value means no and true values means yes.

Note –The align and autostart attributes deprecated in HTML5. Do not use these attributes.

## ...cont'd

- **width:** Width of the object in pixels
- **height:** Height of the object in pixels
- **name:** A name used to reference the object.
- **src:** URL of the object to be embedded.
- **volume:** Controls volume of the sound. Can be from 0 (off) to 100 (full volume).

## ...cont'd

### ✓ Supported Video Types

- ❑ various media types like Flash movies (.swf), AVI's (.avi), and MOV's (.mov) file types inside embed tag.
  - .swf files - created by Macromedia's Flash program.
  - .wmv files - are Microsoft's Window's Media Video file types.
  - .mov files - are Apple's Quick Time Movie format.
  - .mpeg files - created by the Moving Pictures Expert Group.

```
<body>
<embed src="/html/yourfile.swf" width="200" height="200" >
  <noembed></noembed>
</embed>
</body>
```

## ...cont'd

### ✓ Background Audio

- Use HTML <bgsound> tag to play a soundtrack in the background of your webpage
  - this tag is supported by Internet Explorer only and most of the other browsers ignore
  - background sound file also will replay whenever the user refreshes the browser.
  - having only two attributes loop and src.

Note – The bgsound tag is deprecated and it is supposed to be removed in a future version of HTML.

```
<body>
<bgsound src="/html/yourfile.mid">
    <noembed></noembed>
</bgsound>
</body>
```

## ...cont'd

### □ Object tag

- HTML 4 introduces the <object> element, which offers an all-purpose solution to generic object inclusion.
- The <object> element allows HTML authors to specify everything required by an object for its presentation by a user agent.

```
<object data = "data/test.htm" type = "text/html" width = "300" height = "200">  
  alt : <a href = "data/test.htm">test.htm</a>  
</object>
```

```
<object data = "data/test.pdf" type = "application/pdf" width = "300" height = "200">  
  alt : <a href = "data/test.pdf">test.htm</a>  
</object>
```

```
<object data = "data/test.wav" type = "audio/x-wav" width = "200" height = "20">  
  <param name = "src" value = "data/test.wav">  
  <param name = "autoplay" value = "false">  
  <param name = "autoStart" value = "0">  
  alt : <a href = "data/test.wav">test.wav</a>  
</object>
```

# HTML Graphics

---

- ✓ Graphics are representations used to make web-page or applications visually appealing and also for improving user experience and user interaction.
- ✓ Some examples of graphics are photographs, flowcharts, bar graphs, maps, engineering drawings, constructional blueprints, etc.
- ✓ Usually, the following technologies are used in web graphics with HTML5 Canvas API, SVG, PNG, JPG, etc.

## ...cont'd

### □ SVG (Scalable Vector Graphics)

- These are images created by a markup language that are reusable, simple, high-quality standalone images that can be exported and imported as well.
- They are cross-browser friendly and can be used both on the client-side and server-side of the application.
- They can be manipulated to create animations, hybrid images, etc by editing the markup language or by editing using a stylesheet.
- Files come with a .svg extension.
- SVG defines vector-based graphics in XML, which can be directly embedded in HTML pages.
- SVG graphics are scalable, and do not lose any quality if they are zoomed or resized:

```
<svg width="550" height="50" viewBox="0 0 550 50"  
    fill="green" xmlns="http://www.w3.org/2000/svg">  
    <text x="0" y="20">I love GeeksforGeeks</text>  
</svg>
```

## ...cont'd

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<svg width="150" height="100" xmlns="http://www.w3.org/2000/svg">
```

```
  <rect width="100%" height="100%" fill="green" />
```

```
  <circle cx="75" cy="50" r="40" fill="yellow" />
```

```
  <text x="75" y="60" font-size="30" text-anchor="middle" fill="red">SVG</text>
```

```
</svg>
```

```
</body>
```

```
</html>
```

## ...cont'd

### □ PNG (Portable Network Graphics)

- They are portable, static and lossless with proper indexed-color control.
- Files come with a .png or .PNG extension.
- Cross-browser friendly and have streaming capabilities.

`img src="Component1-300x258.png"> </img>`

## ...cont'd

### □ JPG or JPEG (Joint Photographic Experts Group)

- Lossy compressed with an adjustable degree of compression.
- Used mainly with digital photography and can achieve a compression of 10:1.
- Files come with a .jpg or jpeg extension.

```
</img>
```

## ...cont'd

### □ Canvas API:

- Has no DOM and uses vector based methods to create objects, graphics and shapes.
- Canvas API applications can be standalone or integrated with HTML or SVG.
- Widely used for games
- Client-side scripting with native modern browser support.

```
<canvas id="canvas1" width="400" height="200"  
  style="border:2px solid green; border-radius:35px">  
  Your browser does not support the canvas element.  
</canvas>
```



# End of Ch.4

---

Questions, Ambiguities, Doubts, ... ???

